

ESPOO TAPIOLA, Finland

WMO: 029851

Lat: 60.1781N

Lon: 24.7872E

Elev: 8

StdP: 101.23

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.5	-16.4	-22.1	0.5	-19.2	-18.7	0.7	-15.5	10.1	0.6	9.3	0.3	2.8	10	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	26.8	19.4	25.0	18.4	23.3	17.3	20.7	24.9	19.5	23.6	18.3	22.1	3.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	23.0	17.9	12.9	21.4	16.7	11.9	20.2	59.8	25.0	55.5	23.6	51.7	22.1	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.7	6.9	DB	-21.0	28.8	4.6	2.5	-24.3	30.6	-27.0	32.1	-29.5	33.5	-32.9	35.3	(4)
(5)			WB	-21.3	21.4	4.5	1.4	-24.5	22.4	-27.2	23.2	-29.7	24.0	-33.0	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-5.1	-3.7	-0.8	4.6	10.9	14.7	18.1	16.5	12.0	5.8	2.4	-1.2
	DBStd	8.73	5.66	5.56	3.93	3.20	3.95	3.13	3.07	2.68	3.04	3.37	4.06	4.99
	HDD10.0	2113	467	383	335	164	35	3	0	0	16	133	228	348
	HDD18.3	4468	726	617	593	411	232	116	43	70	189	387	478	607
	CDD10.0	743	0	0	0	3	63	144	251	201	77	4	0	0
	CDD18.3	56	0	0	0	0	1	7	36	12	0	0	0	0
	CDH23.3	371	0	0	0	0	14	48	242	67	0	0	0	0
	CDH26.7	56	0	0	0	0	1	4	43	8	0	0	0	0
(14) Wind	WSAvg	3.5	3.6	3.5	3.7	3.3	3.5	3.6	3.2	3.2	3.3	3.6	3.8	4.2
(15) Precipitation	PrecAvg	603	44	32	30	34	36	52	56	73	60	68	66	50
	PrecMax	838	84	100	57	71	73	125	152	165	155	177	170	128
	PrecMin	360	0	0	4	7	5	11	2	1	17	16	4	6
	PrecStd	102	22	23	15	19	20	27	40	40	33	36	31	27
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	6.3	9.8	18.2	25.2	26.8	30.0	27.9	21.1	14.4	10.7	8.6
		MCWB	3.7	3.4	5.1	10.4	16.3	17.4	20.2	20.2	16.4	11.7	9.4	7.6
	2%	DB	3.7	4.1	7.5	15.9	22.6	24.6	27.8	25.2	19.3	12.9	9.4	7.0
		MCWB	3.1	2.8	3.8	8.8	14.4	17.1	19.9	18.9	15.6	11.3	8.7	5.9
	5%	DB	2.8	3.0	5.6	13.0	20.7	22.9	26.0	23.2	18.1	12.0	8.2	5.7
		MCWB	2.4	2.3	2.9	7.9	13.6	16.2	19.3	17.6	15.0	10.8	7.6	4.8
	10%	DB	1.7	2.1	3.8	10.7	18.5	21.0	24.3	21.4	16.9	10.9	7.4	4.5
		MCWB	1.4	1.6	2.2	6.0	12.5	15.0	18.4	16.6	14.5	9.8	6.8	3.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	3.7	6.2	10.9	17.9	20.1	22.0	21.9	17.4	12.8	9.9	8.0
		MCDB	4.3	4.9	8.9	17.3	23.6	24.2	26.8	25.9	19.3	13.7	10.4	8.4
	2%	WB	3.2	3.1	4.4	9.4	15.8	18.2	21.2	19.8	16.5	11.9	8.8	6.2
		MCDB	3.7	3.9	6.4	14.5	20.2	23.2	25.6	23.1	18.4	12.6	9.3	6.8
	5%	WB	2.5	2.4	3.4	8.0	14.6	16.8	20.3	18.7	15.7	11.0	7.7	4.9
		MCDB	2.8	2.9	5.0	12.3	19.0	21.1	24.7	21.9	17.4	11.6	8.2	5.5
	10%	WB	1.4	1.7	2.5	6.7	13.0	15.7	19.1	17.6	14.9	10.0	6.9	3.8
		MCDB	1.7	2.1	3.6	9.9	18.0	20.0	23.4	20.3	16.3	10.7	7.4	4.3
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	4.7	6.4	8.3	9.4	8.5	8.5	8.2	7.0	5.9	3.9	4.1
		MCDBR	3.4	4.5	8.2	12.5	12.6	11.2	10.6	10.2	8.6	7.0	4.2	4.6
		MCWBR	3.2	3.7	5.3	7.0	6.2	5.5	4.5	4.8	5.2	5.3	4.0	4.5
	5% WB	MCDBR	3.4	4.0	7.0	11.5	10.7	10.0	9.5	8.8	7.8	5.6	4.1	4.4
		MCWBR	3.3	3.5	4.9	6.8	5.9	5.4	4.6	4.6	5.2	5.0	4.0	4.2
(40) Clear-Sky Solar Irradiance	taub		0.279	0.313	0.335	0.360	0.358	0.356	0.378	0.372	0.344	0.334	0.307	0.264
	taud		2.139	2.191	2.249	2.344	2.417	2.477	2.438	2.454	2.519	2.394	2.216	2.088
	Ebn at Noon		473	669	786	838	869	876	844	821	786	653	436	333
	Edh at Noon		43	72	94	103	103	99	100	92	73	60	41	30
(44) All-Sky Solar Radiation	RadAvg		0.25	0.82	2.31	3.85	5.25	5.46	5.27	4.08	2.49	1.08	0.30	0.15
	RadStd		0.04	0.16	0.30	0.40	0.57	0.43	0.48	0.42	0.32	0.17	0.05	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	-138	N/A	N/A	N/A

Nomenclature: See separate page